

Publication list - Kento Sato

Latest Publication

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Conference

- HPCAsia 2026 Amarjit Singh, **Kento Sato**, Kohei Yoshida, Kentaro Uesugi, Yasumasa Joti, Takaki Hatsui, Andr  s Rubio Proa  o, ROIX-Comp: Optimizing X-ray Computed Tomography Imaging Strategy for Data Reduction and Reconstruction. In Proceedings of the International Conference on High Performance Computing in Asia-Pacific Region (HPCASIA '26). ACM: Association for Computing Machinery, New York, NY, USA, 2026 [To appear]
- HPCAsia 2026 Ryubu Hosoki, **Kento Sato**, Toshio Endo, Julien Bigot, Edouard Audit, Optimizing Intra-Layer Parallel Communication for LLM Training on Systems with Fully-Connected Mesh GPU Topology. In Proceedings of the International Conference on High Performance Computing in Asia-Pacific Region (HPCASIA '26). ACM: Association for Computing Machinery, New York, NY, USA, 2026 [To appear]
- SoCC 2025 Steven Chien, **Kento Sato**, Artur Podobas, Niclas Jansson, Stefano Markidis, Michio Honda, "ParaLog: Consistent host-side logging for parallel checkpoints", In Proceedings of the 2025 ACM Symposium on Cloud Computing (SoCC '25). Association for Computing Machinery, New York, NY, USA [To appear]
- SC25 Franck Cappello, Robert Underwood, Yuri Alexeev, Alison Baker, Ebru Bozda  , Martin Burtscher, Kyle Chard, Sheng Di, Kyle Gerard Felker, Paul Christopher O'Grady, Hanqi Guo, Yafan Huang, Peng Jiang, Sian Jin, Petter Johansson, Shaomeng Li, Xin Liang, Erik Lindahl, Peter Lindstrom, Zarija Luki  , Magnus Lundborg, Danylo Lykov, Masaru Nagaso, **Kento Sato**, Amarjit Singh, Seung Woo Son, Shihui Song, William Tang, Dingwen Tao, Jiannan Tian, Kazutomo Yoshii, and Kai Zhao. 2025. What to Support When You're Compressing: The State of Practice Gaps and Opportunities for Scientific Data Compression. In Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC '25). Association for Computing Machinery, New York, NY, USA, 1966–1979. <https://doi.org/10.1145/3712285.3759856>
- COOL CHIPS 2025 Tomohiro Ueno, Kaito Kitazume, Masato Kiyama, Kazutomo Yoshii, **Kento Sato**, Norihisa Fujita, Ryohei Kobayashi, Taisuke Boku and Kentaro Sano, "Evaluation of Trade-Off Between Compression Ratio and Hardware Cost for Adaptive Bandwidth Compression Hardware Platform," in 2025 IEEE Symposium on Low-Power and High-Speed Chips and Systems (COOL CHIPS), Tokyo, Japan, 2025, pp. 1-6, doi:10.1109/COOLCHIPS65488.2025.11018561.

- CCGrid 2025 Ryubu Hosoki, **Kento Sato**, Toshio Endo, Julien Bigot and Edouard Audit, “ An Optimization Technique for Hiding Communication Costs in 3D Parallel Training of Deep Learning,” in 2025 IEEE 25th International Symposium on Cluster, Cloud and Internet Computing (CCGrid), Tromsø, Norway, 2025, pp. 1-10, doi: 10.1109/CC-GRID64434.2025.00044. (Best Paper Award)
- PRICAI 2024 Haiwen Chen, Songcan Yu, Shupeng Zhao, Junbo Wang, Kaiming Zhu, **Kento Sato**, “FedSD: Cross-Heterogeneous Federated Learning Based on Self-Distillation” , In 21th Pacific Rim International Conference on Artificial Intelligence, PRICAI 2024, Kyoto, Japan
- SC24 Ana Luisa Veroneze Solorzano, **Kento Sato**, Devesh Tiwari, Keiji Yamamoto, Jim Brandt, Benjamin Schwaller, Sara Petra Walton, Jennifer Green, Fumiyoshi Shoji, “ Toward Sustainable HPC: In-Production Deployment of Incentive-Based Power Efficiency Mechanism on the Fugaku Supercomputer” , SC ’ 24: Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis, Atlanta, GA, USA
- SC24 Xiang Fu, Weiping Zhang, Xin Huang, Wubiao Xu, Shiman Meng, Luanzheng Guo, **Kento Sato**, “ AutoCheck: Automatically Identifying Variables for Checkpointing by Data Dependency Analysis” , SC’24: Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis, Atlanta, GA, USA.
- Cluster 2024 Xiang Fu, Shiman Meng, Weiping Zhang, Luanzheng Guo, **Kento Sato**, Dong H. Ahn, Ignacio Laguna, Gregory L. Lee, Martin Schulz, "Distributed Order Recording Techniques for Efficient Record-and-Replay of Multi-Threaded Programs," in 2024 IEEE International Conference on Cluster Computing (CLUSTER), Kobe, Japan, 2024, pp. 27-38
- ICPADS2023 Andr s Xavier Rubio Proa o, **Kento Sato**, “ Understanding Power Consumption Metric on Heterogeneous Memory Systems” , The 29th IEEE International Conference on Parallel and Distributed Systems (ICPADS 2023), Sentosa, Hainan, China, Dec. 2023
- PDCAT2022 Takaaki Fukai, **Kento Sato** and Takahiro Hirofuchi, “ Analyzing I/O Performance of a Hierarchical HPC Storage System for Distributed Deep Learning” , The 23rd International Conference on Parallel and Distributed Computing, Applications and Technologies (PDCAT ’ 22), December, 2022, Sendai, Japan
- HiPC2021 Akihiro Tabuchi, Koichi Shirahata, Masafumi Yamazaki, Akihiko Kasagi, Takumi Honda, Kouji Kurihara, Kentaro Kawakami, Tsuguchika Tabaru, Naoto Fukumoto, Akiyoshi Kuroda, Takaaki Fukai and **Kento Sato**, “The 16,384-node Parallelism of 3D-CNN Training on An Arm CPU based Supercomputer” , 28th IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC2021), Nov, 2021
- CCGrid2021 Rupak Roy, **Kento Sato**, Subhadeep Bhattacharya, Xingang Fang, Yasumasa Joti, Takaki Hatsui, Toshiyuki Hiraki, Jian Guo and Weikuan Yu, “ Compression of Time Evolutionary Image Data through Predictive Deep Neural Networks” , In the proceedings of the 21 IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGrid 2021), May, 2021

- ICPADS2018 Tianqi Xu, **Kento Sato** and Satoshi Matsuoka, "Explorations of Data Swapping on Burst Buffer", The 24th IEEE International Conference on Parallel and Distributed Systems (ICPADS 2018), Sentosa, Singapore, Dec, 2018. (Acceptance rate: 37.7%, 97/257) (Selected poster presentation: 18.6%, 18/97)
- MASCOT 2018 Yue Zhu, Fahim Chowdhury, Huansong Fu, Adam Moody, Kathryn Mohror, **Kento Sato** and Weikuan Yu, "Entropy-Aware I/O Pipelining for Large-Scale Deep Learning on HPC Systems", IEEE International Symposium on the Modeling, Analysis, and Simulation of Computer and Telecommunication Systems (MASCOTS 2018), Milwaukee, WI, September 2018.
- IPDPS2017 Teng Wang, Adam Moody, Yue Zhu, Kathryn Mohror, **Kento Sato**, Tanzima Islam and Weikuan Yu, "MetaKV: A Key-Value Store for Metadata Management of Distributed Burst Buffers", In Proceedings of the International Conference on Parallel and Distributed Processing Symposium 2017 (IPDPS2017), Orlando, USA, May, 2017.
- PPoPP2017 **Kento Sato**, Dong H. Ahn, Ignacio Laguna, Gregory L. Lee, Martin Schulz and Christopher M. Chabreanu, "Noise Injection Techniques for Reproducing Subtle and Unintended Message Races", Proceedings of the 20th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP17), Austin, USA, Feb, 2017. (Acceptance rate: 22.0%, 29/132)
- ICPADS2016 Tianqi Xu, **Kento Sato** and Satoshi Matsuoka, "CloudBB: Scalable I/O Accelerator for Shared Cloud Storage", The 22nd IEEE International Conference on Parallel and Distributed Systems (ICPADS 2016), Wuhan, China, Dec, 2016. (Acceptance rate: 29.9%, 123/412)
- SC16 Teng Wang, Kathryn Mohror, Adam Moody, **Kento Sato** and Weikuan Yu "An Ephemeral Burst-Buffer File System for Scientific Applications", In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis 2016 (SC16), Salt Lake City, USA, Nov, 2016. (Acceptance rate: 18.4%, 82/446)
- SC15 **Kento Sato**, Dong H. Ahn, Ignacio Laguna, Gregory L. Lee and Martin Schulz, "Clock Delta Compression for Scalable Order-Replay of Non-Deterministic Parallel Applications", In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis 2015 (SC15), Austin, USA, Nov, 2015. (Acceptance rate: 22.1%, 79/358)
- IPDPS2015 Naoto Sasaki, **Kento Sato**, Toshio Endo and Satoshi Matsuoka, "Exploration of Lossy Compression for Application-level Checkpoint/Restart", In Proceedings of the International Conference on Parallel and Distributed Processing Symposium 2015 (IPDPS2015), Hyderabad, INDIA, May, 2015. (Acceptance rate: 21.8%, 108/496)
- CCGrid2014 **Kento Sato**, Kathryn Mohror, Adam Moody, Todd Gamblin, Bronis R. de Supinski, Naoya Maruyama and Satoshi Matsuoka, "A User-level InfiniBand-based File System and Checkpoint Strategy for Burst Buffers", In Proceedings of the 14th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CC-Grid2014), Chicago, USA, May, 2014. (Acceptance rate: 19.1%, 54/283) (**Best Paper Award, 1/54**)

- IPDPS2014 **Kento Sato**, Adam Moody, Kathryn Mohror, Todd Gamblin, Bronis R. de Supinski, Naoya Maruyama and Satoshi Matsuoka, “FMI: Fault Tolerant Messaging Interface for Fast and Transparent Recovery”, In Proceedings of the International Conference on Parallel and Distributed Processing Symposium 2014 (IPDPS2014), Phoenix, USA, May, 2014. (Acceptance rate: 21.1%, 114/541)
- SC12 **Kento Sato**, Adam Moody, Kathryn Mohror, Todd Gamblin, Bronis R. de Supinski, Naoya Maruyama and Satoshi Matsuoka, “Design and Modeling of a Non-blocking Checkpointing System”, In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis 2012 (SC12), Salt Lake, USA, Nov, 2012. (Acceptance rate: 21.2%, 100/472)
- SC12 Akira Nukada, **Kento Sato** and Satoshi Matsuoka, “Scalable Multi-GPU 3-D FFT for TSUBAME 2.0 Supercomputer”, In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis 2012 (SC12), Salt Lake, USA, Nov, 2012. (Acceptance rate: 21.2%, 100/472)
- SC11 Naoya Maruyama, Tatsuo Nomura, **Kento Sato**, and Satoshi Matsuoka, “Physis: An Implicitly Parallel Programming Model for Stencil Computations on Large-Scale GPU-Accelerated Supercomputers”, In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis 2011 (SC11), Seattle, USA, Nov 2011. (Acceptance rate: 21.0%, 74/352)

Journal (Refereed)

- TPDS Kohei Yoshida, Ryuichi Sakamoto, **Kento Sato**, Abhinav Bhatele, Hayato Yamaki, Hiroki Honda, Shinobu Miwa, ” ” VAHRM: Variation-Aware Resource Management in Heterogeneous Supercomputing Systems ” in IEEE Transactions on Parallel & Distributed Systems, vol. , no. 01, pp. 1-15, PrePrints 5555, doi:10.1109/TPDS.2025.3577252.
- IoT-J Xi Zhu, Junbo Wang, **Kento Sato**, Zibin Zheng, “Adaptive Model Compression for Efficient Federated Learning in IoT Systems,” in IEEE Internet of Things Journal, doi: 10.1109/JIOT.2025.3557861
- FGCS Franck Cappello, Sheng Di, Robert Underwood, Dingwen Tao, Jon Calhoun, Yoshii Kazutomo, **Kento Sato**, Amarjit Singh, Luc Giraud, Emmanuel Agullo, Xavier Yepes, Mario Acosta, Sian Jin, Jiannan Tian, Frédéric Vivien, Boyuan Zhang, Kentaro Sano, Tomohiro Ueno, Thomas Grützmacher, Hartwig Anzt, Multifacets of lossy compression for scientific data in the Joint-Laboratory of Extreme Scale Computing, Future Generation Computer Systems, 2024, ISSN 0167-739X
- IEEE TIFS 2023 Taiyu Wang, Qinglin Yang, Kaiming Zhu, Junbo Wang, Chunhua Su, **Kento Sato**, “ LDS-FL: Loss Differential Strategy based Federated Learning for Privacy Preserving, ” in IEEE Transactions on Information Forensics and Security, doi: 10.1109/TIFS.2023.3322328., 2023

- STAM 2023 Satoru Hamamoto, Masaki Oura, Atsuomi Shundo, Daisuke Kawaguchi, Satoru Yamamoto, Hidekazu Takano, Masayuki Uesugi, Akihisa Takeuchi, Takahiro Iwai, Yasuo Seto, Yasumasa Joti, **Kento Sato**, Keiji Tanaka & Takaki Hatsui, Demonstration of efficient transfer learning in segmentation problem in synchrotron radiation X-ray CT data for epoxy resin, *Science and Technology of Advanced Materials: Methods*, DOI: 10.1080/27660400.2023.2270529, 2023 (Selected as Editor's Choice Collection)
- ELSEVIER 2022 Xi Zhu, Junbo Wang, Wuhui Chen, **Kento Sato**, "Model compression and privacy preserving framework for federated learning", *Future Generation Computer Systems*, 2022, ISSN 0167-739X, <https://doi.org/10.1016/j.future.2022.10.026>
- IEEE TBD 2022 Amitangshu Pal, Junbo Wang, Yilang Wu, Krishna Kant, Zhi Liu, **Kento Sato**, "Social Media Driven Big Data Analysis for Disaster Situation Awareness: A Tutorial", in *IEEE Transactions on Big Data*, doi: 10.1109/TBDATA.2022.3158431, Mar., 2022
- IEEE TVT 2022 Feiyuan Liang, Qinglin Yang, Ruiqi Liu, Junbo Wang, **Kento Sato**, Jian Guo, "Semi-Synchronous Federated Learning Protocol with Dynamic Aggregation in Internet of Vehicles," in *IEEE Transactions on Vehicular Technology*, doi: 10.1109/TVT.2022.3148872, Feb., 2022
- IJHPCA 2019 **Kento Sato**, Ignacio Laguna, Gregory L Lee, Martin Schulz, Christopher M Chamberau, Simone Atzeni, Michael Bentley, Ganesh Gopalakrishnan, Zvonimir Rakamaric, Geof Sawaya, Joachim Protze, and Dong H Ahn. 2019. Pruners: Providing reproducibility for uncovering non-deterministic errors in runs on supercomputers. *Int. J. High Perform. Comput. Appl.* 33, 5 (Sep 2019), 777–783. DOI:<https://doi.org/10.1177/1094342019834621>
- IJHPCA 2019 Chapp, D., Rorabaugh, D., **Kento Sato**, Ahn, D. H., & Taufer, M. (2019). A three-phase workflow for general and expressive representations of nondeterminism in HPC applications. *The International Journal of High Performance Computing Applications*, 33(6), 1175–1184.
- JSFI 2018 Dylan Chapp, **Kento Sato**, Dong H. Ahn and Michela Taufer, "Record-and-Replay Techniques for HPC Systems: A Survey", *Supercomputing Frontiers and Innovations* (180102) 2018

Workshops (Refereed)

- IWAHPCE 2026 Yosuke Asai, **Kento Sato**, Keiji Yamamoto, Hitoshi Murai and Kohei Yoshida, "System Software Utilization on an ARM-Based Supercomputer: Insights from a Production-Scale System", In *Proceedings of the 2026 International Conference on High Performance Computing in Asia-Pacific Region Workshops (HPC Asia '26 Workshops)*, , Association for Computing Machinery, New York, NY, USA (IWAHPCE2026: International Workshop on Arm-based HPC: Practice and Experience 2026) [To appear]

- EE HPC SOP 2026 Kohei Yoshida, Hayato Yamaki, Hiroki Honda, **Kento Sato**, Shinobu Miwa, “Combining System- and User-Level Approaches to Improving Energy Efficiency in GPU-Based Supercomputers”, In Proceedings of the 2026 International Conference on High Performance Computing in Asia-Pacific Region Workshops (HPC Asia ’26 Workshops). Association for Computing Machinery, New York, NY, USA, 1–10. (EE HPC SOP: Energy Efficient HPC State of the Practice) [To appear]
- LLM4HPCAsia 2026 Jian Guo, Jianwen Wei, Yufei Cheng, Jiajie Sheng, Yijun Wu and **Kento Sato**, “Runtime Prediction for Local Deployment of Large Language Models: A Case Study on Qwen Models Covering LoRA Fine-Tuning, RAG, and Inference”, In Proceedings of the 2026 International Conference on High Performance Computing in Asia-Pacific Region Workshops (HPC Asia ’26 Workshops). Association for Computing Machinery, New York, NY, USA, [To appear]
- PDSW 2025 Minqiu Sun, Xin Huang, Luanzheng Guo, Nathan R. Tallent, **Kento Sato**, and Dong Dai. 2025. LLMTailor: A Layer-wise Tailoring Tool for Efficient Checkpointing of Large Language Models. In Proceedings of the SC ’25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC Workshops ’25). Association for Computing Machinery, New York, NY, USA, 1366–1374. <https://doi.org/10.1145/3731599.3767515> (PDSW ’25: The 10th International Parallel Data Systems Workshop)
- EPSOUQ-HPC 2024 (in SC24) Xin Huang, Weiping Zhang, Shiman Meng, Wubiao Xu, Xiang Fu, Luanzheng Guo, **Kento Sato**, “Scrutinizing Variables for Checkpoint Using Automatic Differentiation”, The Second Workshop on Enabling Predictive Science with Optimization and Uncertainty Quantification in HPC (EPSOUQ-HPC) in conjunction with SC24
- IXPUG 2024 (in SC24) Wubiao Xu, Xin Huang, Shiman Meng, Weiping Zhang, Luanzheng Guo and **Kento Sato**, “An Efficient Checkpointing System for Large Machine Learning Model Training”, The Intel eXtreme Performance Users Group (IXPUG) Workshop in conjunction with SC24
- ESSA 2024 (in IPDPS2024) Xiang Fu, Xin Huang, Wubiao Xu, Shiman Meng, Weiping Zhang, Luanzheng Guo, **Kento Sato**, “Benchmarking variables for checkpointing in HPC Applications”, The 5th Workshop on Extreme-Scale Storage and Analysis (ESSA2024) in 2023 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW), June, San Francisco, USA , 2024
- APDCM 2023 (in IPDPS2023) Fu Xiang, Hao Tang, Huimin Liao, Xin Huang, Wubiao Xu, Shimeng Meng, Weiping Zhang, Luanzheng Guo and **Kento Sato**, “A High-dimensional Algorithm-Based Fault Tolerance Scheme,” APDCM 2023, IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW), St. Petersburg, Florida USA, 2023, DOI: 10.1109/IPDPSW59300.2023.00061

- MLHPC 2021 (in SC21) Steven Farrell, Murali Emani, Jacob Balma, Lukas Drescher, Aleksandr Drozd, Andreas Fink, Geoffrey Fox, David Kanter, Thorsten Kurth, Peter Mattson, Dawei Mu, Amit Ruhela, **Kento Sato**, Koichi Shirahata, Tsuguchika Tabaru, Aristeidis Tsaris, Jan Balewski, Ben Cumming, Takumi Danjo, Jens Domke, Takaaki Fukai, Naoto Fukumoto, Tatsuya Fukushima, Balazs Gerofi, Takumi Honda, Toshiyuki Imaura, Akihiko Kasagi, Kentaro Kawakami, Shuhei Kudo, Akiyoshi Kuroda, Maxime Martinasso, Satoshi Matsuoka, Kazuki Minami, Prabhat Ram, Takashi Sawada, Mallikarjun Shankar, Tom St. John, Akihiro Tabuchi, Venkatram Vishwanath, Mohamed Wahib, Masafumi Yamazaki, Junqi Yin and Henrique Mendonca, “MLPerf HPC: A Holistic Benchmark Suite for Scientific Machine Learning on HPC Systems”, The Workshop on Machine Learning in High Performance Computing Environments (MLHPC) 2021 in conjunction with SC21, Nov, 2021
- HPS'20 (in IPDPS2020) Tonmoy Dey, **Kento Sato**, Bogdan Nicolae, Jian Guo, Jens Domke, Weikuan Yu, Franck Cappello, and Kathryn Mohror. 2020. “Optimizing Asynchronous Multi-Level Checkpoint/Restart Configurations with Machine Learning.” The IEEE International Workshop on High-Performance Storage, May, 2020
- ROSS 2018 (in HPDC 2018) Yue Zhu, Teng Wang, Kathryn Mohror, Adam Moody, **Kento Sato**, Muhib Khan and Weikuan Yu, “Direct-FUSE: Removing the Middleman for High-Performance FUSE File System Support”, The 8th International Workshop on Runtime and Operating Systems for Supercomputers (ROSS 2018) in conjunction with the 27th International Symposium on High-Performance Parallel and Distributed Computing (HPDC 2018), Tempe, Arizona, USA, 2018.
- CRE2017 (in SC17) **Kento Sato**, Ignacio Laguna, Gregory L. Lee, Martin Schulz, Christopher M. Chamberau, Dong H. Ahn, Simone Atzeni, Michael Bentley, Genesh Gopalakrishnan, Zvonimir Rakamaric, Geof Sawaya and Joachim Protze, “PRUNERS: Providing Reproducibility for Uncovering Non-deterministic Errors in Runs on Supercomputers” (WIP talk), Computational Reproducibility at Exascale (CRE) at Supercomputing 2017 (SC17), Denver, USA, Nov, 2017.
- PDSW-DISCS 2017 (in SC17) Yue Zhu, Teng Wang, Kathryn Mohror, Adam Moody, **Kento Sato**, Muhib Khan and Weikuan Yu, “Direct-FUSE: Removing the Middleman for High-Performance FUSE File System Support” (WIP talk), 2nd Joint International Workshop on Parallel Data Storage and data Intensive Scalable Computing Systems (PDSW-DISCS) at Supercomputing 2017 (SC17), Denver, USA, Nov, 2017.
- Workshop (in ICDCS2017) Kevin Brown, Tianqi Xu, Keita Iwabuchi, **Kento Sato**, Adam Moody, Kathryn Mohror, Nikhil Jain, Abhinav Bhatele, Martin Schulz, Roger Pearce, Maya Gokhale and Satoshi Matsuoka, "Accelerating Big Data Infrastructure and Applications (Ongoing collaboration)", The 1st US-Japan Workshop on Collaborative Global Research on Applying Information Technology in conjunction with The 37th IEEE International Conference on Distributed Computing Systems (ICDCS 2017), Atlanta, USA, June, 2017
- ExaMPI15 (in SC15) Aiman Fang, Ignacio Laguna, **Kento Sato**, Tanzima Islam and Kathryn Mohror, “Fault Tolerance Assistant (FTA): An Exception Handling Approach for MPI Programs” (Hot topic), Workshop on Exascale MPI (ExaMPI15) at Supercomputing 2015 (SC15), Austin, USA, Nov, 2015.

- EPForDM2015 Moshe Gabel, **Kento Sato**, Daniel Keren, Satoshi Matsuoka and Assaf Schuster, (in EDBT) “Latent Fault Detection With Unbalanced Workloads”, Event Processing, Forecasting and Decision-Making in the Big Data Era 2015 (EPForDM2015) in conjunction with the 18th International Conference on Extending Database Technology (EDBT), Brussels, Belgium, March, 2015.
- FTXS2013 Takafumi Saito, **Kento Sato**, Hitoshi Sato and Satoshi Matsuoka, “Energy-aware I/O Optimization for Checkpoint and Restart on a NAND Flash Memory System”, In the Workshop on Fault-Tolerance for HPC at Extreme Scale 2013 (FTXS2013) in conjunction with the International Symposium on High Performance Parallel and Distributed Computing (HPDC13), New York, USA, June, 2013.
- Cloud2009 **Kento Sato**, Hitoshi Sato and Satoshi Matsuoka, “A Model-Based Algorithm for Optimizing I/O Intensive Applications in Clouds using VM-Based Migration”, In the International Workshop On Cloud Computing (Cloud2009) in conjunction with the International Symposium on Cluster Computing and the Grid 2009 (CCGrid2009), Shanghai, China, May, 2009.

Poster (Refereed)

- HPCAsia 2026 Joanna Ayulia Benyamin, Ke Cui, Amarjit Singh, **Kento Sato**, “Enhancing Scientific Image Prediction and Compression through AI Model Fine-Tuning”, In Proceedings of the International Conference on High Performance Computing in Asia-Pacific Region (HPCASIA ’26). Association for Computing Machinery, New York, NY, USA, (Poster) [To appear]
- HPDC 2025 Mina Atef Yousef, Amarjit Singh, **Kento Sato**, “Refactoring TEZip: Integrating Python-Based Predictive Compression into an HPC C++/LibTorch Environment”, In Proceedings of the 34th International Symposium on High-Performance Parallel and Distributed Computing (HPDC ’25), July, 2025
- R-CCSIS’26 Kohei Yoshida, **Kento Sato**, Shinobu Miwa, Rio Yokota, “Evaluating the Impact of Batch Independent Normalization and Shuffling in DDP”, the 7th R-CCS International Symposium (RCCS-IS7), Kobe, Japan, Jan. 2025
- SC24 Alexis Amoyo, Amarjit Singh, **Kento Sato**, Weikuan Yu, “Exploration of Super Resolution Techniques for Image Compression”, In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis 2024 (SC24), Research Posters, Atlanta, USA, Nov, 2024
- SC24 Akshay Nambudiripad, Amarjit Singh, **Kento Sato**, Weikuan Yu, “Development of TEZip in PyTorch: Integrating New Prediction Models into an Existing Compression Framework”, In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis 2024 (SC24), Research Posters, Atlanta, USA, Nov, 2024
- R-CCSIS’24 Andr  s Rubio Proa  o and **Kento Sato**, “Power Consumption Metric on Heterogeneous Memory Systems”, the 6th R-CCS International Symposium (RCCS-IS6), Kobe, Japan, Feb. 2024
- R-CCSIS’24 Amarjit Singh and **Kento Sato**, “Research and Development of an Infrastructure for Big Data Collection, Analysis, and Utilization in Large Scale Research Facilities”, the 6th R-CCS International Symposium (RCCS-IS6), Kobe, Japan, Feb. 2024

- SC23 Isita Talukdar, Amarjit Singh, Robert Underwood, **Kento Sato**, Weikuan Yu, “Integrating TEZIP into LibPressio: A Case Study of Integrating a Dynamic Application into a Static C Environment”, In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis 2023 (SC23), Research Poster, Denver, USA, Nov, 2023 (Poster).
- APNet’23 Steven W. D. Chien, **Kento Sato**, Artur Podobas, Niclas Jansson, Stefano Markidis and Michio Honda, “Improving Cloud Storage Network Bandwidth Utilization of Scientific Applications”, 7th Asia-Pacific Workshop on Networking, June, 2023
- ISC23 Amarjit Singh and **Kento Sato**, "Development of an Infrastructure for Big Data Collection, Analysis, and Utilization in Large Scale Research Facilities", 9th IEEE International Smart Cities Conference 2023 (ISC23), May, 2023
- R-CCSIS’23 Andr  s Rubio Proa  o and **Kento Sato**, “Power Consumption Metric on Heterogeneous Memory Systems”, the 5th R-CCS International Symposium (RCCS-IS5), Kobe, Japan, Feb. 2023
- R-CCSIS’23 Amarjit Singh and **Kento Sato**, “Research and Development of an Infrastructure for Big Data Collection, Analysis, and Utilization in Large Scale Research Facilities”, the 5th R-CCS International Symposium (RCCS-IS5), Kobe, Japan, Feb. 2023
- R-CCSIS’22 Takaki Fukai, **Kento Sato** “Measurement of I/O Performance on a Hierarchical File System for Distributed Deep Neural Network”, the 4th R-CCS International Symposium (RCCS-IS4), Kobe, Japan, Feb. 2021 (Lightning Presentation)
- R-CCSIS’21 Takaaki Fukai, **Kento Sato**, “Measurement of I/O performance for distributed deep neural networks on Fugaku”, The 3rd R-CCS International Symposium, Feb, 2021
- SC19 Tonmoy Dey, **Kento Sato**, Jian Guo, Bogdan Nicolae, Jens Domke, Weikuan Yu, Franck Cappello and Kathryn Mohror, “Optimizing Asynchronous Multi-level Checkpoint/Restart Configurations with Machine Learning”, In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis 2019 (SC19), Regular Poster, Denver, USA, Nov, 2019.
- SC19 Rupak Roy, **Kento Sato**, Jian Guo, Jens Domke and Weikuan Yu, “Improving Data Compression with Deep Predictive Neural Network for Time Evolutional Data”, In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis 2019 (SC19), Regular Poster, Denver, USA, Nov, 2019.
- SC18 Yue Zhu, Fahim Chowdhury, Huansong Fu, Adam Moody, Kathryn Mohror, **Kento Sato** and Weikuan Yu, “Multi-Client DeepIO for Large-Scale Deep Learning on HPC Systems”, In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis 2018 (SC18), Regular Poster, Dallas, USA, Nov, 2018. (Acceptance rate: 56%, 93/165)
- ISC 2018 Tianqi Xu, **Kento Sato** and Satoshi Matsuoka, “HuronFS: Hierarchical, User-level and On-demand Burst Buffer File System”, In Proceedings of the International Supercomputing Conference (ISC 2018), Frankfurt, Germany, USA, June, 2018

- SC17 Tianqi Xu, **Kento Sato** and Satoshi Matsuoka, “ A Simulation-Based Analysis on the Configuration of the Burst Buffer”, In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis 2017 (SC17), Regular Poster, Denver, USA, Nov, 2017. (Acceptance rate: 58%, 98/169) (**Best Poster Finalist, 9/98**)
- SC17 Dylan Chapp, **Kento Sato**, Dong H. Ahn and Michela Taufer, “Towards Capturing Nondeterminism Motifs in HPC Applications”, In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis 2017 (SC17), ACM Student Research Competition Poster, Denver, USA, Nov, 2017. (Acceptance rate: 47%, 28/59)
- SC15 Tianqi Xu, **Kento Sato** and Satoshi Matsuoka, “Design and Modelling of Cloud-based Burst Buffers”, In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis 2015 (SC15), Austin, USA, Nov, 2015. (Acceptance rate: 44%, 112/253)
- SC15 Teng Wang, Kathryn Mohror, Adam Moody, Weikuan Yu and **Kento Sato**, “BurstFS: A Distributed Burst Buffer File System for Scientific Applications”, In Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis 2015 (SC15), Austin, USA, Nov, 2015. (Acceptance rate: 44%, 112/253)
- ISC'15 Tianqi Xu, **Kento Sato** and Satoshi Matsuoka, “Towards Cloud-based Burst Buffers for I/O Intensive Computing in Cloud”, In HPC in Asia Workshop in conjunction with the International Supercomputing Conference (ISC'15), Frankfurt, Germany, July, 2015.
- ISC'14 Naoto Sasaki, **Kento Sato**, Toshio Endo and Satoshi Matsuoka, “Exploration of Application-level Lossy Compression for Fast Checkpoint/Restart”, In HPC in Asia Workshop in conjunction with the International Supercomputing Conference (ISC'14), Leipzig, Germany, June, 2014.
- GTC2014 **Kento Sato**, Akira Nukada, Naoya Maruyama and Satoshi Matsuoka , “I/O acceleration with GPU for I/O-bound Applications”, GPU Technology Conference 2014, Mar. 2014.
- ISC'12 **Kento Sato**, Adam Moody, Kathryn Mohror, Todd Gamblin, Bronis R. de Supinski, Naoya Maruyama and Satoshi Matsuoka, “Towards a Light-weight Non-blocking Checkpointing System”, In HPC in Asia Workshop in conjunction with the International Supercomputing Conference (ISC'12), Hamburg, Germany, June, 2012.
- Grid2008 **Kento Sato**, Hitoshi Sato and Satoshi Matsuoka, “Model-based Optimization for Data-Intensive Application on Virtual Cluster”, In the International Conference on Grid Computing (Grid 2008), Tsukuba, Sep, 2008.
- SACIS2008 佐藤 賢斗, 佐藤仁, 松岡聡, “仮想クラスタを用いた Data-Intensive Application 実行環境の性能モデル構築と最適化に向けて”, 先進的計算基盤システムシンポジウム (SACIS2008), Tsukuba, June, 2008.

Workshops (Non-refereed)

- Tech Report 2024 Ryubu Hosoki, **Kento Sato**, Toshio Endo, Julien Bigot, Edouard Audit. “An optimization pass for training speed-up and strategy search in 3D parallelism”, IPSJ SIG Technical Report, 2024-HPC-194, No.7, May 8, 2024
- Tech report 2021 深井 貴明, 広瀬 崇宏, 高野 了成, Akram Ben Ahmed, **佐藤 賢斗**, “FPGA による次世代メモリのエミュレーション機構”, 第 180 回 研究報告ハイパフォーマンスコМПユーティング (HPC 研究会), July 2021.
- Tech report 2020 Atsushi Nukariya, Kazutoshi Akao, Jin Takahashi, Naoto Fukumoto, Kentaro Kawakami, Akiyoshi Kuroda, Kazuo Minami, **Kento Sato** and Satoshi Matsuoka, “HPC and AI Initiatives for Supercomputer Fugaku and Future Prospects”, Fujitsu Technical Review, November, 2020
- Tech report 2015 Tianqi Xu, **Kento Sato** and Satoshi Matsuoka, “Cloud-based Burst Buffers for I/O Acceleration”, IPSJ SIG Technical Reports 2015-HPC-150, Beppu, Japan, Aug, 2015.
- Tech report 2014 Tianqi Xu, **Kento Sato** and Satoshi Matsuoka, “Towards Cloud Bursting for Extreme Scale Supercomputers”, IPSJ SIG Technical Reports 2014-HPC-145, Nigata, Jul, 2014.
- Tech report 2014 佐々木尚人, **佐藤 賢斗**, 遠藤敏夫, 松岡聡, “実アプリケーションにおけるウェーブレット変換を用いたチェックポイントデータの非可逆圧縮手法”, IPSJ SIG Technical Reports 2014-HPC-145, Nigata, Jul, 2014
- Tech report 2013 **Kento Sato**, Satoshi Matsuoka, Adam Moody, Kathryn Mohror, Todd Gamblin, Bronis R. de Supinski and Naoya Maruyama, “Burst SSD Buffer: Checkpoint Strategy at Extreme Scale”, IPSJ SIG Technical Reports 2013-HPC-141, Okinawa, Sep, 2013.
- Tech report 2013 松岡 聡, **佐藤 賢斗**, 遠藤敏夫 “エクサスケールスパコンに向けた耐故障性の評価 — TSUBAME2.0 を例にして —”, IPSJ SIG Technical Reports 2013-HPC-141, Okinawa, Sep, 2013.
- SWoPP 2012 **Kento Sato**, Adam Moody, Kathryn Mohror, Todd Gamblin, Bronis R. de Supinski, Naoya Maruyama and Satoshi Matsuoka, “Design and Modeling of an Asynchronous Checkpointing System”, IPSJ SIG Technical Reports 2012-HPC-135 (SWoPP 2012), Tottori, Aug, 2012.
- HOKKE-19 **Kento Sato**, Adam Moody, Kathryn Mohror, Todd Gamblin, Bronis R. de Supinski, Naoya Maruyama and Satoshi Matsuoka, “Towards an Asynchronous Checkpointing System”, IPSJ SIG Technical Reports 2011-ARC-197 2011-HPC-132 (HOKKE-19), Hokkaido, Nov, 2011.
- SWoPP2008 **佐藤 賢斗**, 佐藤仁, 松岡聡, “仮想クラスタを用いた Data-Intensive Application 実行環境の性能モデル構築と最適化”, 並列／分散／協調処理に関するサマワークショップ (SWoPP2008), Saga, Aug, 2008.

Poster (Non-refereed)

- 2012 **Kento Sato**, Adam Moody, Kathryn Mohror, Todd Gamblin, Bronis R. de Supinski, Naoya Maruyama and Satoshi Matsuoka, “Design and Modeling of a Non-Blocking Checkpoint System”, In ATIP - A*CRC Workshop on Accelerator Technologies in High Performance Computing, Singapore, March, 2012.

- 2011 **Kento Sato**, Hitoshi Sato and Satoshi Matsuoka “Orchestrated Data Processing Acceleration for Data-Intensive Applications by using VM-based Migration”, The 1st Data Intensive Science Workshop, Tokyo, March, 2011.

Preprint Archive

- arXiv 2025 Du Wu, Enzhi Zhang, Isaac Lyngaas, Xiao Wang, Amir Ziabari, Tao Luo, Peng Chen, **Kento Sato**, Fumiyoshi Shoji, Takaki Hatsui, Kentaro Uesugi, Akira Seo, Yasuhito Sakai, Toshio Endo, Tetsuya Ishikawa, Satoshi Matsuoka, Mohamed Wahib, “Paradigm Shift in Infrastructure Inspection Technology: Leveraging High-performance Imaging and Advanced AI Analytics to Inspect Road Infrastructure.” arXiv preprint arXiv:2505.13955 (2025).
- arXiv 2025 Franck Cappello and Allison Baker and Ebru Bozda and Martin Burtcher and Kyle Chard and Sheng Di and Paul Christopher O Grady and Peng Jiang and Shaomeng Li and Erik Lindahl and Peter Lindstrom and Magnus Lundborg and Kai Zhao and Xin Liang and Masaru Nagaso and **Kento Sato** and Amarjit Singh and Seung Woo Son and Dingwen Tao and Jiannan Tian and Robert Underwood and Kazutomo Yoshii and Danylo Lykov and Yuri Alexeev and Kyle Gerard Felker, “Lossy Compression of Scientific Data: Applications Constrains and Requirements”, March, 2025, arXiv:2503.20031
- arXiv 2024 Steven W. D. Chien, **Kento Sato**, Artur Podobas, Niclas Jansson, Stefano Markidis and Michio Honda, “Accelerating Scientific Application through Transparent I/O Interposition”, arXiv/2401.14576, Jan., 2024
- arXiv 2023 Takahiro Hirofuchi, Takaaki Fukai, Akram Ben Ahmed, Ryousei Takano and **Kento Sato**, “METICULOUS: An FPGA-based Main Memory Emulator for System Software Studies”, Sep., 2023, arXiv:2309.06565

Invited Talks and Panels

- 2025 佐藤 賢斗, AI for Science 時代の大規模データ処理ソフトウェア技術, 第4回 HPK-理研研究交流会 (2025 年 12 月 23 日-24 日)
- 2025 **Kento Sato**, Software Solutions for Large-scale Data Management in Scientific Research, LSSDA: Productive, Performant Software for Large Scale Scientific Data Analysis October 22, 2025, 8:30–8:50am (Session 5)
- 2025 BoF: Co-Designing Next Generation Supercomputing Systems in ISC25
- 2025 佐藤 賢斗, 「スパコンを研究するとは？」(富岳 富岳 NEXT 編), 神戸大学 システム情報学部 - システム情報学入門 (2025 年 5 月 29 日)
- 2025 佐藤 賢斗, 富岳 NEXT × AI: 次世代スーパーコンピュータが拓く科学の未来, 第 69 回 システム制御情報学会 研究発表講演会 (SCI ' 25) - TS13 機械学習と制御 – 招待講演 (2025 年 5 月 27 日)
- 2025 **Kento Sato**, Fugaku and Beyond: Cloud-Enabled Supercomputer bridging Europe and Japan, EU JAPAN DIGITAL WEEK 2025, March 31, 2025
- 2024 佐藤 賢斗, “富岳 NEXT がもたらす科学の革新: AI for Science”, データサイエンスの研究と教育の将来: 「富岳」NEXT の展望, 兵庫県立大学大学院情報科学研究科, 2024 年 12 月 26 日

- 2024 佐藤 賢斗, “「富岳」、「富岳 Next」でシンカする AI — 国産 LLM 構築技術のしくみと AI for Science”, 第 6 回 中高生のためのオンライン特別授業 「今、神戸でアツい科学」, 2024 年 11 月 2 日
- 2024 佐藤 賢斗, “AI for Science のためのシステムソフトウェアと次世代計算基盤”, CEATEC2024 - AI for All 特設ステージ, 2024 年 10 月 18 日
- 2024 Kento Sato, “Data in HPC: Optimization, Compression and Analysis”, REX-IO24: The 4th Workshop on Re-envisioning Extreme-Scale I/O for Emerging Hybrid HPC Workload”, September 24, 2024
- 2024 Kento Sato, “Feasibility Study on System software for FugakuNext – Status update”, InPEX: The International Post-Exascale Project 2024, June 17 – 19, 2024
- 2024 Kento Sato, “TEZip: Data data compression tool for time evolutionary data”, FZ Workshop, February 14-15, 2024
- 2024 Kento Sato, “Feasibility study on FugakuNext system software and the survey of software usage on Fugaku”, ADAC: Portability, Sustainability and Integration WG Seminar, February 8, 2024
- 2023 Kento Sato, “System Software Research – The FugakuNext Supercomputer and Synchrotron Radiation Facilities”, WSSP36 (36th Workshop on Sustained Simulation Performance), December 11, 2023
- 2023 佐藤 賢斗, “スーパーコンピュータがもたらす AI の革命”, 2023 年度スパコンセミナー覗いてみようスパコンの世界 - 「富岳」を見て、聞いて、知って, October 14, 2023
- 2023 佐藤 賢斗, “大型放射光施設のデータ処理を支えるシステムソフトウェア技術”, PCC ワークショップ in 大阪 2023, June 22, 2023
- 2023 Kento Sato, "Feasibility Study at the System Software and Library Research Group", The 5th R-CCS International Symposium, February 6, 2023
- 2023 佐藤 賢斗, “「次世代計算基盤に係る調査研究」理化学研究所チーム システムソフト G の取組”, 科学技術計算分科会 2022 年度会合「富岳」NEXT への挑戦 現在から未来へ, January 20, 2023
- 2022 佐藤 賢斗, “「富岳」を中心とした大規模データ処理システム”, 情報処理学会 連続セミナー 2022 第 12 回 「「富岳」とスパコン技術の展望」, December 13, 2022
- 2022 Kento Sato, ReMPI: A Record-and-Replay Tool for Debugging Non-Deterministic MPI Applications, SuperCheck@SC22, November 14, 2022
- 2022 Kento Sato, “The Supercomputer Fugaku - AI and Big Data”, A Workshop on Modeling Materials at Realistic time Scales via Optimal Exploitation of Exascale Computers and Artificial Intelligence, July 25, 2022
- 2022 佐藤 賢斗, “スーパーコンピュータ「富岳」 - 世界最速機械学習システム -”, 第 27 回 AT 研究会オープンアカデミックセッション (ATOS27), January 21, 2022
- 2021 佐藤 賢斗, “富岳と大型施設との連携に向けて”, 第 1 回 SPring-8 データワークショップ (SPring-8 データセンター構想におけるデータ解析能力の強化), December 13, 2021

- 2021 佐藤 賢斗, “ High Performance Big Data Systems for Extreme-scale Data Science on Fugaku ”, International Workshop on the Integration of Simulation + Data + Learning: Towards Society5.0 by h3-Open-BDEC, November 30, 2021
- 2021 佐藤 賢斗, “ スーパーコンピュータ「富岳」 - 世界最速級機械学習システム - ”, 第4回機械学習工学研究会 (MLSE 夏合宿 2021) , June 2, 2021 (Online)
- 2020 Kento Sato, “High Performance System Software Enabling Convergence of AI, Big data and HPC”, ADAC9 Workshop, September, 2020
- 2020 Kento Sato, “Convergence of AI/BD and HPC - Data compression and DL4Fugaku”, Inauguration Meeting of Synchrotron for Neuroscience – an Asia Pacific Strategic Enterprise (SYNAPSE), Singapore, January, 2020)
- 2019 佐藤 賢斗, “「富岳」AI 利用の展望”, In 第14回 AI サービス研究会, Tokyo, December, 2019.
- 2019 佐藤 賢斗, “高性能ビッグデータ処理とポストムーア時代に向けたアプリケーション解析”, In JACORN (Japan Consortium for the Reconfigurable-hardware Next generation), October, 2019.
- 2019 Kento Sato, “Convergence of AI/Big data and HPC”, In 4th International Symposium on Research and Education of Computational Science (RECS), October, 2019.

Books

- 2023 Satoshi Matsuoka, Kento Sato, Mohamed Wahib and Aleksandr Drozd, “The First Exascale Supercomputer Accelerating AI-for-Science and Beyond”, Artificial Intelligence for Science, Chapter 9, 145-161, DOI:10.1142/9789811265679_0009, May, 2023.
- 2013 Satoshi Matsuoka, Takayuki Aoki, Toshio Endo, Hitoshi Sato, Shin'ichiro Takizawa, Akihiko Nomura and Kento Sato, “TSUBAME2.0: The First Petascale Supercomputer in Japan and the Greatest Production in the World”, volume 1, chapter 20, pages 525-556. Chapman & Hall/CRC Computational Science, April 2013. URL <http://www.crcnetbase.com/doi/book/10.1201/b14677>.

Dissertations

- Doctoral Degree Kento Sato, “Design and Implementatin for Optimal Checkpoint/Restart”, Tokyo Institute of Technology, Committee members (Prof. Satoshi Matsuoka, Prof. Toshio Endo, Prof. Hidehiko Masuhara, Prof. Naoto Miyoshi and Prof. Ken Wakita), 2014.
- Master Degree Kento Sato, “仮想マシンマイグレーションを考慮した大規模データ処理の最適化”, Tokyo Institute of Technology, Commitemembers (Prof. Masataka Sassa, (2010) Prof. Osamu Watanabe, Prof. Shigeru Chiba, Prof. Ken Wakita and Prof. Satoshi Matsuoka), 2010.
- Bachelor Degree Kento Sato, “仮想クラスタを用いたデータインテンシブアプリケーション実行環境の性能モデル構築と最適化”, Tokyo Institute of Technology, Advisor (Satoshi (2008) Matsuoak), 2008.